

BREASTGUARD AI PLATFORM

Early Alert Breast Cancer Detection & Risk Intelligence Platform

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Report ID: f22ad23f-84ee

Facility: General Oncology Research Laboratory

Patient Name: Sarah Miller

Sex: Female

Age: 58

Patient ID: BG-2026-0002

Address: Sabha

AI Diagnosis Prediction:

Confidence Score:

Calculated Probability:

Breast Cancer Detected

82.27%

82.27% (HIGH RISK)

AI Diagnostic Explanation (Explainable SHAP analysis)

The prediction is mainly influenced by CEA, CA-15.3, LDH and ESR values which are strongly associated with breast cancer progression.

Key Contributing Parameter	Measured Patient Value	Reference Range	Clinical Status	Probability Impact
CEA (Carcinoembryonic Antigen)	8.40 ng/mL	0.0-3.0 ng/mL	Elevated	+16.6%
CA-15.3 (Cancer Antigen 15-3)	55.20 U/mL	0.0-30.0 U/mL	Elevated	+13.9%
FBS (Fasting Blood Sugar)	110.00 mg/dL	70.0-100.0 mg/dL	Elevated	-5.1%
LDH (Lactate Dehydrogenase)	290.00 U/L	100.0-250.0 U/L	Elevated	+2.7%
Age (Age)	58 years	N/A	Normal	+2.5%

Breast Cancer Future Risk Projections (Trend Timeline)

Assessment Interval	Current Point	1 Month	3 Months	6 Months	1 Year
Risk Probability	82.27%	99.77%	100.0%	100.0%	100.0%

Clinical Recommendations & Referrals

- Immediate oncology consultation advised.
- Additional imaging tests (Ultrasound/MRI) recommended.
- Biopsy may be requested by the oncologist.
- Consult Oncologist.



QR Verification Code

Complete Clinical Lab Panel

Lab Parameter	Measured Value	Healthy Reference Limits	Status
ALB (Albumin)	3.40 g/dL	3.5-5.0 g/dL	Decreased
ALP (Alkaline Phosphatase)	135.00 U/L	44.0-147.0 U/L	Normal
CA-15.3 (Cancer Antigen 15-3)	55.20 U/mL	0.0-30.0 U/mL	Elevated
CEA (Carcinoembryonic Antigen)	8.40 ng/mL	0.0-3.0 ng/mL	Elevated
CL (Chloride)	98.00 mEq/L	96.0-106.0 mEq/L	Normal
Creatinin (Creatinine)	1.10 mg/dL	0.5-1.2 mg/dL	Normal
ESR (Erythrocyte Sedimentation Rate)	35.00 mm/hr	0.0-20.0 mm/hr	Elevated
FBS (Fasting Blood Sugar)	110.00 mg/dL	70.0-100.0 mg/dL	Elevated
GOT (AST (GOT))	34.00 U/L	10.0-40.0 U/L	Normal
GPT (ALT (GPT))	42.00 U/L	7.0-56.0 U/L	Normal
HGB (Hemoglobin)	11.80 g/dL	12.0-16.0 g/dL	Decreased
K (Potassium)	3.80 mEq/L	3.5-5.0 mEq/L	Normal
LDH (Lactate Dehydrogenase)	290.00 U/L	100.0-250.0 U/L	Elevated
Na (Sodium)	137.00 mEq/L	135.0-145.0 mEq/L	Normal
PLT (Platelets)	390.00 x10 ³ /uL	150.0-450.0 x10 ³ /uL	Normal
RBC (Red Blood Cells)	4.00 x10 ⁶ /uL	4.0-5.4 x10 ⁶ /uL	Normal
T_Ca (Total Calcium)	9.60 mg/dL	8.5-10.2 mg/dL	Normal
Urea (Blood Urea)	35.00 mg/dL	15.0-45.0 mg/dL	Normal
WBC (White Blood Cells)	9.80 x10 ³ /uL	4.0-11.0 x10 ³ /uL	Normal